



SEMESTER 2 EXAMINATIONS 2018/2019

MODULE: CA216 - Operating Systems

PROGRAMME(S):

CASE BSc in Computer Applications (Sft.Eng.)
ECSAO Study Abroad (Engineering & Computing)

YEAR OF STUDY: 2,O

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

Please use the files provided to answer the questions.

You are allowed to refer to materials (one A4 page) which you have brought into the examination venue.

Please only use the files provided for each question, and save your work regularly. Whilst answer booklets may be used for rough work, you must use the files provided to answer questions.

Please include your exam number on each file in the space provided.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1

[TOTAL MARKS: 40]

Please open the file named **q1.txt** and use it to answer the questions below. Simply write the answer (A-E) in the space provided. All questions carry equal marks (4%) and no negative marking applies.

Q 1(a)

[4 Marks]

A **foreground process** is one which:

- A. Has total control of the operating system's resources.
- B. Has a low priority.
- C. Runs permanently.
- D. Must be started by the user.
- E. Is a process which is currently open and appearing on the display, and with keyboard or other I/O directed to it.

Q 1(b)

[4 Marks]

The **kernel mode** of an Operating System is defined as:

- A. One in which all instructions are available, and in which only the kernel runs.
- B. A process which has total control of all other processes.
- C. One in which only single threaded applications can run.
- D. A mode in which user programs always run.
- E. None of the above.

Q 1(c)

[4 Marks]

The word **livelock** is defined as:

- A. Another word for deadlock.
- B. A situation where every process in the system is running perfectly.
- C. A situation where a thread continuously attempts an action which fails.
- D. A situation where no file can be accessed.
- E. A situation where two or more threads have higher priority than all other threads in the system, and as a consequence cause the other threads to experience starvation.

Q 1(d)

[4 Marks]

In the context of **memory management**, the term **compaction** means:

- A. Compressing files so that they occupy as little space as possible.
- B. The rearrangement of information stored, to leave one or more holes, in order to make storage more efficient.
- C. Moving data around to leave one very big hole at the end of available memory.
- D. Removing all data from memory, then reallocating the space more efficiently.
- E. Completely filling available memory with no gaps or holes left whatsoever.

Q 1(e)

[4 Marks]

Small, fast memory located between the processor and main memory is called:

- A. Block memory
- B. Cache memory
- C. Direct memory
- D. WORM memory
- E. Read Only Memory

Q 1(f)

[4 Marks]

A **Race Condition** is a situation where:

- A. Multiple threads or processes are cooperating to complete a task.
- B. A situation where a thread cannot complete its task because it has run out of time.
- C. A situation where two threads are concurrently trying to change the value of a variable.
- D. A situation where a thread is quickly terminated.
- E. A situation where the CPU must allocate resources to a process or thread within a certain amount of time.

Q 1(g)

[4 Marks]

In the context of Operating Systems, a **pipe** is:

- A. A logical conduit allowing two processes to communicate.
- B. A construct allowing one process to create another.
- C. A link between main and cache memory.
- D. A link between I/O devices and the CPU.
- E. A mechanism for allocating memory to a list.

Q 1(h)

[4 Marks]

An **asynchronous thread** is one which:

- A. Cannot be created.
- B. A threading model in which a parent creates a child thread, then resumes execution thereby allowing both threads to execute concurrently and independently.
- C. A threading model which enables the parent thread to create a child, but which also means that the parent must wait until the child thread is finished before it can resume execution.
- D. Does not have a critical section.
- E. A threading model which means that at most two threads may be created.

Q 1(i)

[4 Marks]

Which of the following is the unit of work in a system?

- A. A process.
- B. A semaphore
- C. A programme.
- D. A monitor.
- E. A mode bit

Q 1(j)

[4 Marks]

A buffer with no practical limit on its size is known as:

- A. A bounded buffer.
- B. A limited buffer.
- C. An infinite buffer.
- D. An unbounded buffer.
- E. A max Buffer.

[End of Question1]

QUESTION 2**[TOTAL MARKS: 20]**

Please open the file named **q2.txt** and use it to answer the following questions.

Q 2(a)**[8 Marks]**

What is meant by the term **semaphore**. Explain the different types of semaphore, and the operations which may be performed upon them. Include details in your answer of when, and why, you would use each type.

Q 2(b)**[8 Marks]**

The Dining Philosopher's problem can be prone to situations where processes can be **starved**. Explain what is meant by the term starved. Outline a solution to avoid this. You should include code snippets to illustrate your answer. Note that you may use either Python or Pseudocode.

Q 2(c)**[4 Marks]**

Explain what is meant by the term **monitor**. **Note that two sentences here will suffice.**

[End of Question2]

QUESTION 3**[TOTAL MARKS: 20]**

Please open the file named **q3.txt** and use it to answer the following questions.

Q 3(a)**[10 Marks]**

What is meant by the term process? Also explain what is meant by the term thread. Give 3 differences between a process and a thread.

Q 3(b)**[10 Marks]**

A process may be in any one of five states. List these states, and give a one-sentence explanation for each.

[End of Question3]

QUESTION 4 **[TOTAL MARKS: 20]**

Please open the file named **q4.txt** and use it to answer the following questions.

Q 4(a) **[8 Marks]**

Explain what is meant by the terms **sequential access** and **random access** as they are used in the context of files. Give an example of when, and where, each might be used.

Q 4(b) **[8 Marks]**

Some systems provide file sharing by maintaining a single copy of the file. Other systems maintain several copies, one for each of the users who are sharing the file. Briefly describe the benefits of each approach. Also, give one drawback to each way of doing things.

Q 4(c) **[4 Marks]**

In the **Linux Operating System** a file might have permissions set to **RWXR--R--** (or **OCTAL 744**). What do these permissions mean?

[End of Question4]

[END OF EXAM]